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Housing market connectedness and transmission of monetary policy

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Abstract

This paper investigates whether interconnectivity among local housing markets influences the effectiveness of the (United States) U.S. monetary transmission mechanism. We construct measures of housing market connectedness and employ a state-dependent local projection method to estimate nonlinear impulse responses of macroeconomic variables to monetary policy (MP) shocks. The findings show that MP has a greater impact when regional housing markets are more synchronized. Higher interconnectivity amplifies spillover effects across local markets, enhancing MP effectiveness. Additionally, we find that MP is particularly effective when highly connected states experience economic expansion.

KEYWORDS
housing market connectedness, monetary policy, state-dependent model

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R31, E52, E32

1 | INTRODUCTION

U.S. regional housing markets are synchronized, with varying levels of synchronization over time (Choi & Andrew Hansz, 2021; Landier et al., 2017). Such interconnectivity among local housing markets has the potential to play a crucial role in the monetary transmission mechanism because changes in regional-level economic conditions triggered by monetary policy shocks could spill over to other regions. Since the spillover effect may amplify or dampen the effects of monetary policy, it remains unclear whether monetary policy is more or less effective when regional housing markets are more synchronized. This presents a challenge for policymakers seeking to tailor monetary policy to achieve desired economic outcomes. Against this backdrop, this paper aims to examine the impact of housing market connectedness on the monetary transmission mechanism in the U.S. economy.

Abbreviations: FFR, federal funds rate; FMHPI, Freddie Mac House Price Index; GDP, gross domestic product; HP, Hodrick–Prescott; LASSO, least absolute shrinkage and selection operator; MP, monetary policy; PCE, personal consumption expenditures; VAR, vector autoregression; VIX, volatility index; ZLB, zero lower bound.

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To begin with, we measure the degree of interconnectivity among state-level housing markets by adopting the methodology developed by Diebold and Yilmaz (2014). This involves measuring the connectedness index through the variance decomposition matrix associated with an N-variable vector autoregression (VAR). The methodology is useful as it quantifies how shocks in one part of the housing market *spill over* into others. Specifically, it captures weighted, directed networks that characterize connectedness within the VAR system, reflecting both the intensity and direction of the spillovers. Additionally, Diebold and Yilmaz (2014)'s method allows for dynamic analysis, tracking how housing market connectedness evolves over time.

We use data on the average house prices from the Freddie Mac House Price Index (FMHPI) for all 50 states and Washington, D.C. in the United States, covering the period from 1976:m1 to 2020:m12. Based on the connectedness measures, we identify which periods constitute the low- or high-connectedness state.[1] We then employ state-dependent local projection methods, proposed by Jorda (2005), to empirically document whether the transmission of monetary shocks depends on the housing market connectedness. Our empirical methodology is comparable to that of Tenreyro and Thwaites (2016) or Alpanda and Zubairy (2019), as they investigate how monetary policy influences economies in times of recession or when households have high debt levels. As such, our study contributes to the existing literature on the effects of monetary shocks, which can vary depending on the economic conditions.

On the one hand, high connectedness in housing markets can enhance monetary policy effectiveness by amplifying changes through credit and collateral channels. On the other hand, when local economies are highly connected, they lose insulation from external shocks, negatively affecting policy effectiveness.[2] Empirical literature presents mixed findings on whether integration or synchronization among local economies amplifies or dampens the effects of macroeconomic shock propagation. Some studies suggest that synchronization dampens the effectiveness of monetary policy (Arcabic & Skrinjaric, 2021; Bruno & Shin, 2014). However, other research, such as Georgiadis and Mehl (2016), indicates that greater connectedness among local economies amplifies the impact of monetary policy. Our analysis shows that federal funds rate (FFR) shocks have a stronger effect in states with high levels of housing market connectedness compared to those with lower levels of connectedness. More specifically, the impact of a monetary shock on Gross Domestic Product (GDP), consumption, residential investment, house prices, and employment is substantially greater when the initial level of housing market connectedness is higher than its trend. These results are robust to employing different definitions for our state variable, alternative monetary policy shocks, and various specifications for housing market connectedness variables. In particular, our baseline estimation is designed to avoid the impact of the Great Recession and the Zero Lower Bound (ZLB) period, and thus, we limit the sample period to run from 1981:q1 to 2007:q4. We later expand our sample period and discover that our findings hold up even when recent periods are included. Hence, our results appear to be robust to the inclusion of additional data.

Upon analyzing the effects of monetary shocks on states that intersect both business cycles and housing market connectedness cycles, we discovered an intriguing finding: monetary policy is most potent when high-connectedness states are undergoing an expansion. Furthermore, we investigated the impact of monetary policy shocks on house prices across different local housing markets and how the degree of market connectedness influences these dynamics. Our findings suggest that local house prices tend to move together and increase in response to an expansionary monetary policy shock when regional housing markets are more synchronized. In contrast, the response of local house prices is muted and more diverse in low-connectedness states. The results indicate that the existence of a national factor that reflects the synchronized behavior of all housing markets could be essential for enhancing the effectiveness of monetary policy in high-connectedness states.

The significance of this paper lies in providing insight into whether the spillover effect among local housing markets intensifies or mitigates the effects of monetary policy. Changes in interest rates have the potential to affect housing markets in several ways. One of these ways is by lowering mortgage rates, which can make it easier for potential homebuyers to finance their purchases. The increased demand for housing in one region can spill over to other regions, further increasing aggregate demand. Our findings suggest that this effect can be even more pronounced in synchronized housing markets.

Our paper contributes to the extensive body of research that investigates how the state of the economy affects the effectiveness of monetary policy. The literature provides mixed evidence regarding the effectiveness of monetary policy during different phases of the business cycle. Weise (1999) and Lo and Piger (2005) suggest that the impact of money supply shocks on output is stronger during recessions or periods of low output growth than during expansions, while Tenreyro and Thwaites (2016) find that the response of U.S. output to monetary policy shocks is stronger during expansions due to larger responses of consumer durables and business investment expenditures during booms.

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In recent years, there has been a growing focus on how leverage affects the effectiveness of monetary policy. Jorda, Schularick and Taylor (2020) and Alpanda and Zubairy (2019) find that the effects of monetary policy are greater during credit booms and weaker during periods of high household debt, respectively. Using state-dependent local projection methods and data from 18 advanced economies, Alpanda et al. (2021) also show that the impact of monetary policy shocks on output and other macroeconomic and financial variables is weaker during economic downturns, low household debt, and high interest rates. Concurrently, Cloyne et al. (2019) thoroughly assess the impacts of monetary policy by utilizing survey micro data from both the U.S. and the U.K. They categorize households based on their housing tenure status and unveil that the aggregate response of consumption to monetary policy shocks is primarily driven by households that hold mortgages. In a similar vein, using U.K. loan-level micro-data, Cumming and Hubert (2021) investigate how the distribution of the household debt-to-income ratio influences the impact of monetary policy on aggregate consumption. Their main finding is that the response of consumption to monetary policy is strengthened when the portion of households with substantial debt is significant. Their subsequent analysis in Cumming and Hubert (2022) shows that the effect of monetary policy on consumption is more pronounced when house prices have recently decreased.

While the existing literature extensively examines how the state of the economy affects the effectiveness of monetary policy, little research has been conducted on the relationship between regional markets and monetary policy. Previous studies have examined the regional effects of U.S. monetary policy (Furceri et al., 2019), the influence of the regional distribution of housing equity on the consequences of monetary policy (Beraja et al., 2018), and the integration of national monetary and regional housing markets via the mortgage rate (Fratantoni & Schuh, 2003). Using a dynamic factor model and state-level data from 1986 to 2005, Del Negro and Otrok (2007) find that movements in house prices are mainly driven by the local component. In contrast to these studies, our paper focuses specifically on the role of interconnectedness among local housing markets in shaping the effectiveness of monetary policy.[3] Chen et al. (2023) explore the effectiveness of incorporating connectivity information to forecast house price growth rates. Hurn et al. (2022) study patterns in housing market networks using Australian and Chinese data, finding that inter-temporal connectivity strengthens over time, with sentiment playing varying roles across different networks.

The paper is organized as follows: Section 2 outlines the empirical methodology and dataset used in this study. In Section 3, we present the key findings, including local housing market dynamics, the relationship between connectedness and the national factor, and the relationship between connectedness and the business cycle. Section 4 provides a discussion of these findings. We perform a sensitivity analysis in Section 5, and finally, in Section 6, we offer concluding remarks and suggestions for future research.

2 | ECONOMETRIC METHODOLOGY

This section provides a brief overview of the empirical approach used to study how the degree of connectedness among housing markets affects the transmission of monetary policy. To this end, we take two steps. First, we follow the Diebold and Yilmaz (2014) approach to measure the level of housing market linkage. Second, we use state-dependent local projection methods to empirically document whether the transmission of monetary shocks depends on housing market connectedness.

2.1 | Measuring housing market connectedness

To measure the housing market connectedness index, we employ the connectedness approach proposed by Diebold and Yilmaz (2014). According to this approach, the connectedness index can be computed using the variance decomposition matrix associated with an N-variable vector autoregression (VAR). Consider a covariance stationary N-variable $VAR(p)$:

$$X_t = \sum_{m=1}^{p} \Phi_m X_{t-m} + \varepsilon_t,$$

where $\varepsilon_t \sim (0, \Sigma)$. The moving average (MA) representation can be written as:

$$X_t = \sum_{m=0}^{\infty} A_m \varepsilon_{t-m},$$

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where the $N \times N$ coefficient matrices A_m follows the recursion:

$$A_m = \Phi_1 A_{m-1} + \Phi_2 A_{m-2} + .. + \Phi_p A_{m-p},$$

with $A_0 = I_N$ and $A_m = 0$ for $m < 0$. In VAR model estimation, the number of parameters to be estimated increases with the number of variables, which is so-called *curse of dimensionality*. To address this problem, we adopt the elastic net estimator developed by Zou and Hastie (2005), which is a hybrid of shrinkage and selection methods. The elastic net estimator, $\widehat{\beta}_{Enet}$, solves the following:

$$\widehat{\beta}_{Enet} = \arg\min_{\beta}\left[\sum_{t=1}^{T}\left(y_t - \sum_{k=1}^{N}\beta_k x_{kt}\right)^2 + \lambda\sum_{k=1}^{N}\left(\alpha|\beta_k| + (1-\alpha)\beta_k^2\right)\right].$$

Elastic net estimator combines Least Absolute Shrinkage and Selection Operator (LASSO) penalty and Ridge penalty. There are has two tuning parameters, λ and $\alpha \in [0, 1]$. If $\alpha = 1$, it becomes LASSO penalty, and if $\alpha = 0$, it becomes Ridge regression. In this paper, we follow Bostanci and Yilmaz (2020) and set α to be equal to 0.5 without cross validation,[4] while λ is chosen by 10-fold cross validation.

Variance decomposition, which includes coefficients in the MA representation and transformations such as impulse response functions, is crucial in understanding dynamics. This is because the variance decomposition matrix enables us to evaluate the portion of future uncertainty in market i that is caused by shocks in market j, for all $i \neq j$. Identification of forecast error variance decomposition is one of the critical steps when measuring connectedness. In this regard, we utilize the generalized approach proposed by Pesaran and Shin (1998), which is independent of variable ordering and allows for correlated shocks.[5] We will first define highly granular pairwise directional connectedness. The contribution of market j to the Q-step-ahead generalized forecast error variance of market i, $\theta_{ij}^g(Q)$, for $Q = 1, 2, ...$, is defined as:

$$\theta_{ij}^g(Q) = \frac{\sigma_{jj}^{-1}\sum_{h=0}^{Q-1}\left(e_i' A_h \Sigma e_j\right)^2}{\sum_{h=0}^{Q-1}\left(e_i' A_h \Sigma A_h' e_i\right)},$$

where σ_{jj} is the standard deviation of the disturbance of the jth equation, and e_l is a selection vector in which only the l-th element is one and zeros otherwise. $\theta_{ij}^g(Q)$ is interpreted as *cross variance shares*: the fractions of the Q-step-ahead error variances in forecasting market i due to shocks to market j. Having chosen the generalized identification method for the variance decompositions, the sum of each row may not necessarily equal one (i.e., $\sum_{j=1}^{N}\theta_{ij}^g(Q) \neq 1$) since the shocks to each market are not orthogonalized. Taking this into account, we normalize each entry of the generalized variance decomposition matrix by its respective row sum. Pairwise directional connectedness from market j to market i, $C_{ij}(Q)$, is defined as:

$$C_{ij}(Q) = \frac{\theta_{ij}^g(Q)}{\sum_{j=1}^{N}\theta_{ij}^g(Q)}.$$

By construction, $\sum_{j=1}^{N} C_{ij}(Q) = 1$ and $\sum_{i,j=1}^{N} C_{ij}(Q) = N$. Finally, we obtain the *system-wide* connectedness, $\mathcal{C}$, by summing all non-diagonal entries of the normalized variance decomposition matrix:

$$\mathcal{C} = \frac{\sum_{i,j=1,i\neq j}^{N} C_{ij}(Q)}{\sum_{i,j=1}^{N} C_{ij}^H(Q)} = \frac{1}{N}\sum_{i,j=1,i\neq j}^{N} C_{ij}(Q). \tag{1}$$

Accordingly, this total connectedness is an index that measures the interdependence of the entire housing market and ranges from 0 to 1.

2.2 | Local projection method

Our econometric model closely resembles the methodology employed in Ramey and Zubairy (2018). We apply the local projection technique proposed by Jorda (2005) to estimate impulse responses to exogenous monetary policy innovations

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in both linear state-dependent models. The impulse response of variable y_t at horizon $h \in \{0, 1, 2, ...\}$ to the identified monetary shock, ε_t, is estimated as the coefficient β_h in the following linear model:

$$y_{t+h} = \alpha_h + \boldsymbol{\gamma}'\boldsymbol{x}_t + \beta_h \varepsilon_t + u_{t+h}, \tag{2}$$

where is α_h is a constant, and $\boldsymbol{x}_t$ is a vector of control variables. The coefficient β_h is the response of y at time $t + h$ to the shock at time t.[6] Thus, the local projection method simply requires estimation of a series of regressions for each horizon, h, and for each variable of interest.

The linear projection model can be easily adapted to estimate a non-linear, state-dependent model. For the model that allows state-dependence, we estimate a set of regressions for each horizon h as follows:

$$y_{t+h} = D_{H,t-1}\left[\alpha_{H,h} + \boldsymbol{\gamma_{H,h}}'\boldsymbol{x}_t + \beta_{H,h}\varepsilon_t\right] + D_{L,t-1}\left[\alpha_{L,h} + \boldsymbol{\gamma_{L,h}}'\boldsymbol{x}_t + \beta_{L,h}\varepsilon_t\right] + u_{t+h}, \tag{3}$$

where $D_{s,t-1}$ is a dummy variable that indicates the state of the economy in terms of the housing market connectedness before the monetary policy shock hits, $s \in \{L, H\}$.[7] For example, $D_{L,t-1}$ takes a value of one in the low-connectedness state and 0 otherwise. We will discuss how we construct this dummy variable in more detail in the next subsection. It should be noted that all the coefficients other than the deterministic trends in the non-linear model vary depending on the state of the economy.

2.3 | Defining low- and high-connectedness states

In order to study whether monetary policy effectiveness varies according to the housing market connectedness, we need to identify which periods constitute the low- or high-connectedness state. We base our state variable on the *system-wide* connectedness, $\mathcal{C}_t$, in Equation (1). We construct a measure for the housing market connectedness using the FMHPI, which is a measure of typical monthly price inflation for houses across all 50 states and Washington, D.C. in the U.S.[8] The sample period used for the connectedness index runs from January 1976 to December 2020. The connectedness index is computed using an Elastic net LASSO-VAR(1) with a rolling-window analysis, where the forecast horizon is 10 months and the rolling window size is 120 months[9] We then convert our monthly measures to quarterly ones. Next, we define the states of the economy in terms of housing market connectedness as a deviation from a smooth trend. This detrended measure is constructed based on a Hodrick and Prescott (1997) filter (HP filter) with the usual smoothing parameter for quarterly data, that is, $\lambda = 1,600$. We use sample periods from Q1 1981 to Q4 2007 in the baseline estimation.[10]

Figure 1 presents the level and trend of the connectedness index for the baseline sample period (1981:q1 to 2007:q4). We identify a period as a high-connectedness state (low-connectedness state) if the deviation from the trend in that period is positive (negative). It should be noted that 53% of the sample consists of positive connectedness gaps (i.e., high-connectedness states). The identification of low- or high-connectedness episodes does not necessarily coincide with business cycles, as the housing market connectedness does not exhibit a cyclical behavior. Therefore, the correlation between housing market connectedness and output fluctuations is close to zero (−0.08).[11]

2.4 | Identifying monetary policy shocks

In order to estimate impulse response functions using the local projection methods outlined above, it is necessary to specify the assumptions necessary for identifying the monetary policy shocks. As our baseline identification, we follow Alpanda and Zubairy (2019) and consider a standard identification approach employed in a structural VAR model with timing restrictions where GDP and inflation are ordered before the FFR. Accordingly, the key identifying assumption in our baseline estimation is that contemporaneous GDP and inflation are included in the information set of the central bank. To this end, we include contemporaneous and lagged values of GDP and inflation in Equations (2) and (3) along with contemporaneous federal funds rates as the shock and the lagged values of the FFR as part of controls.[12] This is equivalent to the aforementioned identification in the three-variable VAR model. Given the well-known serial

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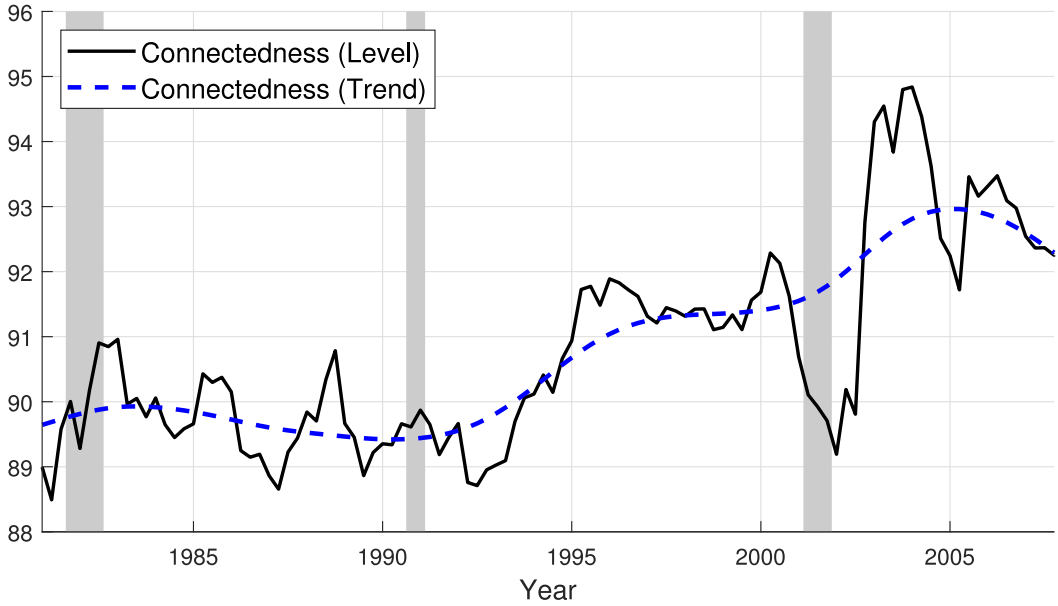

FIGURE 1 Connectedness index: Trend and cyclical components. The trend and level of the connectedness index. The gray-shaded regions indicate NBER recession periods.

correlation problem induced by the successive leading of the dependent variable in the Jorda's method, we apply the correction method proposed in Newey and West (1987) to our standard errors.

As a robustness check, we will also consider the monetary policy shocks developed by Romer and Romer (2004). However, we use the FFR as our benchmark shock measure, particularly because in Section 5.2, we test whether the main result is robust when including the ZLB period, during which Romer and Romer (2004) shocks are unavailable. This allows for a fair comparison, using the FFR as the consistent benchmark specification.

3 | EMPIRICAL RESULTS

In this section of the paper, we report results from the local projection model on the state-dependent effects of monetary policy shocks. We first present our baseline results, where we identify the monetary policy shock using a Choleski identification scheme (or timing restrictions). We explore whether our baseline results are robust to alternative identification of monetary policy shocks following Romer and Romer (2004). We then examine the robustness of our findings by conducting several checks based on sample choice and the definition of the state and threshold.

3.1 | Baseline results

Figure 2 displays the impulse responses of GDP, the Personal Consumption Expenditures (PCE) deflator, and the federal funds rate (FFR) to a 100 basis point expansionary shock for our baseline specification. In the first column, the point estimates of the impulse responses for the three models are shown: high connectedness (lines with triangles), low connectedness (lines with squares), and a linear model (lines with circles). In the second column, the impulse response and 68% confidence intervals for the linear model are displayed, while the third column shows the impulse response functions and 68% confidence intervals for the high- and low-connectedness states to a monetary shock.

We will first discuss the results from the linear model shown in the second column of 2. The linear model presents a familiar picture. Following an expansionary monetary shock, (i) the level of output begins to increase, with its response peaking at around 0.5% above baseline between 16 and 18 quarters after impact, (ii) the price level shows a sticky initial response for the first few quarters but eventually increases by 0.8%, and (iii) the response of the FFR is negative on impact.

Regarding the state-dependent impulse responses, the third column of Figure 2 clearly shows a sharp difference between high- and low-connectedness episodes. The responses of GDP and the price level are significantly larger in the

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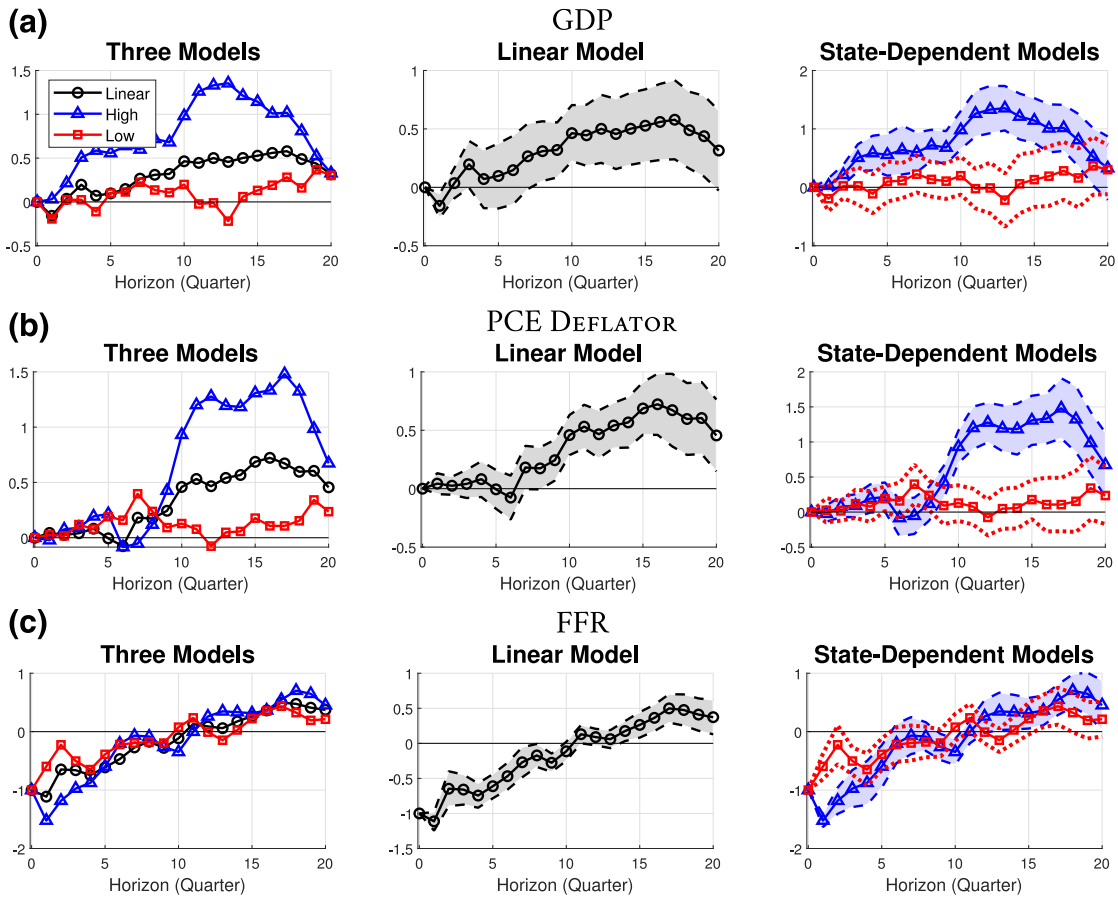

FIGURE 2 Impulse response of headline variables: Baseline specification. The impulse responses of GDP, the personal consumer expenditure (PCE) deflator, and the federal funds rate (FFR) to a 100 basis points (bps) expansionary shock for our baseline specification. The first column displays the point estimate of the impulse response of the three models: high connectedness (lines with triangles), low connectedness (lines with squares), and a linear model (lines with circles). The second column shows the impulse response and 68 percent confidence intervals for the linear model, while the third column shows the impulse response functions and 68 percent confidence intervals for the high- and low-connectedness states.

high-connectedness state. To be more specific, the GDP response peaks at around 0.7% in response to a 100 bps shock in the high-connectedness state, which is larger than the 0.5% in the linear model. The price level responds more strongly in the high-connectedness state than in the linear model, with the maximum rise of around 1.5% in the high-connectedness state. In the low-connectedness state, on the contrary, the responses of GDP and the PCE deflator are not significantly different from zero at most horizons. These results may suggest that there might be stronger demand effects in the high-connectedness state than in the low-connectedness state. It should be noted that the responses of the FFR in both states are similar at most horizons (particularly for horizons after 4 quarters), which implies that the larger responses of GDP and the price level in the high-connectedness state, particularly at medium-run horizons (between 10 and 15 quarters), are not attributable to a bigger fall in the FFR.

In Table 1, we formally test whether the cumulative impulse response functions for the level of GDP and inflation in the high-connectedness state are statistically different from those in the low-connectedness state. The results show that the cumulative effects on the level of GDP and inflation are significantly larger at standard levels, particularly at longer horizons. The third panel of Table 1 verifies that the cumulative response of the FFR is not statistically different between high- and low-connectedness states at most horizons.

Figure 3 depicts the responses of different components of GDP to the same monetary policy shock mentioned earlier. Specifically, we consider private consumption (nondurable goods and services) and fixed investment (including both residential and nonresidential fixed investment) as two important expenditure aggregates.[13] Similar to the response of aggregate output, the two expenditure aggregates show a much stronger response in the high-connectedness state compared to the linear model. However, in the low-connectedness state, the responses of the two expenditure

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TABLE 1 Statistical significance: Baseline specification.

	Cumulative impact at horizon				
	$h = 4$	$h = 8$	$h = 12$	$h = 16$	$h = 20$
GDP					
High connectedness	1.3383	3.8518	8.1047	12.8231	15.5004
Low connectedness	−0.2570	0.3166	0.5868	0.7497	1.8609
p-value	0.2336	0.2425	0.0725	0.0403	0.0657
PCE deflator					
High connectedness	0.3369	0.5327	4.3676	9.3854	13.8484
Low connectedness	0.2397	1.2247	1.4454	1.8355	2.6728
p-value	0.9085	0.7109	0.2856	0.0148	0.0062
FFR					
High connectedness	−5.5547	−6.5049	−6.8548	−5.4948	−3.1462
Low connectedness	−2.9722	−3.9528	−3.8434	−3.3828	−2.2157
p-value	0.0672	0.4166	0.4354	0.5588	0.7616

Note: The cumulative impulse response functions for the level of GDP, inflation, and FFR, and the *p*-value for the null hypothesis that the cumulative response in the high-connectedness state is equal to that in the low-connectedness state at a given horizon.

variables are statistically insignificant at most horizons. We also focus on the response of residential fixed investment, shown in Panel (C) of Figure 3, which is closely related to the housing market connectedness. As expected, residential investment exhibits a larger response in the high-connectedness state.

Figure 4 displays the impulse responses of three other macroeconomic variables that can aid in understanding the state-dependent effects of monetary policy shocks. Panel (A) reveals strong evidence that employment responds significantly more in the high-connectedness state, which is consistent with the GDP response. Furthermore, as shown in Panel (B), the real wage responds more strongly in the high-connectedness state than in the low-connectedness state at most horizons. This could be another piece of evidence for stronger demand effects in the high-connectedness state, where a larger increase in demand prompts firms to hire more workers, leading to a greater rise in the real wage rate, as predicted by standard New Keynesian models. The larger demand effect in the high-connectedness state could be attributed to an increase in real house prices through a home equity channel, which operates by relaxing the borrowing constraint as house prices increase. According to Panel (C), the response of real house prices to a monetary shock is significantly greater in the low-debt state than in the high-debt state. This suggests that if a housing market boom is associated with the high-connectedness state, then monetary policy could be more effective in that state.

Our findings are significant in shedding light on whether spillover effects among neighboring housing markets intensify or mitigate the impact of monetary policy. Changes in interest rates have the potential to affect the housing market in various ways, such as lowering mortgage rates and making it easier for potential homebuyers to finance their purchases. Additionally, increased demand for housing in one region can spill over to other regions, resulting in a further increase in aggregate demand. Our study suggests that this effect can be particularly pronounced in synchronized housing markets.

3.2 | Alternative policy shocks: Romer and Romer (2004)

The monetary policy shocks used in the baseline specification are identified as the structural shocks recovered from a three-variable VAR (GDP, the PCE deflator, and the FFR), using Choleski orthogonalization with the FFR placed last. Alternatively, we use the extended series for the Romer and Romer (2004) monetary shockwhere measures are obtained as the residuals from an estimated reaction function. The residuals obtained are exogenous with respect to the evolution of economic activity in principle. Using the Romer and Romer (2004) monetary shock, we conduct the same analysis as in the previous subsection.

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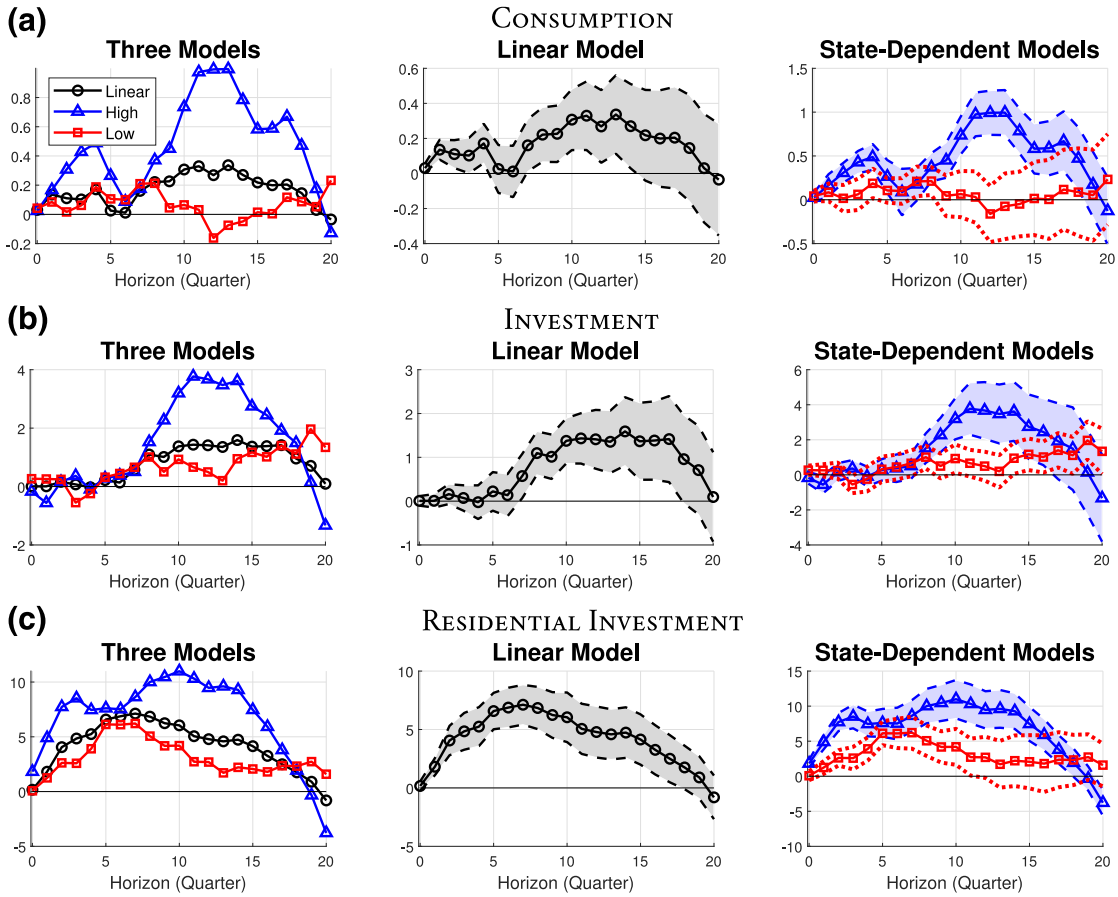

FIGURE 3 Impulse response of expenditure variables: Baseline specification. The impulse responses of consumption (nondurable goods and services), private fixed investment, and residential fixed investment to a 100 basis points (bps) expansionary shock for our baseline specification. The first column displays the point estimate of the impulse response of the three models: high connectedness (lines with triangles), low connectedness (lines with squares), and a linear model (lines with circles). The second column displays the impulse response and 68 percent confidence intervals for the linear model, while the third column shows the impulse response functions 68 percent confidence intervals to a monetary shock for the high- and low-connectedness states.

Figure 5 illustrates the impulse response functions of the key variables (GDP, the PCE deflator, and the FFR) to a Romer and Romer (2004) monetary policy shock. In the linear model, as shown in the second column, the level of output starts to increase and reaches a maximum 4 years after the shock. The price level is initially sticky and starts to increase with a delay. The FFR initially falls and reverts toward the conditional mean. The state-dependent effects of the monetary shock on our key variables are reported in the third column of Figure 5. Once again, GDP and inflation have larger responses in the high-connectedness state. Overall, there is very little difference from our baseline results in terms of qualitative perspectives.[14]

Figure 6 displays the impulse response functions of consumption, investment, and residential investment. Although the differences between the two states become much smaller, there is still evidence suggesting a larger effect of monetary policy shocks on key expenditure variables, particularly on residential fixed investment. Figure 7 presents the effects of monetary shocks on three other macroeconomic variables in the two states. There is still evidence that the responses of employment, real wages, and real house prices are greater in the high-connectedness state, which may indicate stronger aggregate demand effects in that state.

To summarize, our baseline results are robust to shocks identified in the manner of Romer and Romer (2004). Qualitatively, the results retain the message that monetary policy is statistically more expansionary in the high-connectedness state than in the low-connectedness state.

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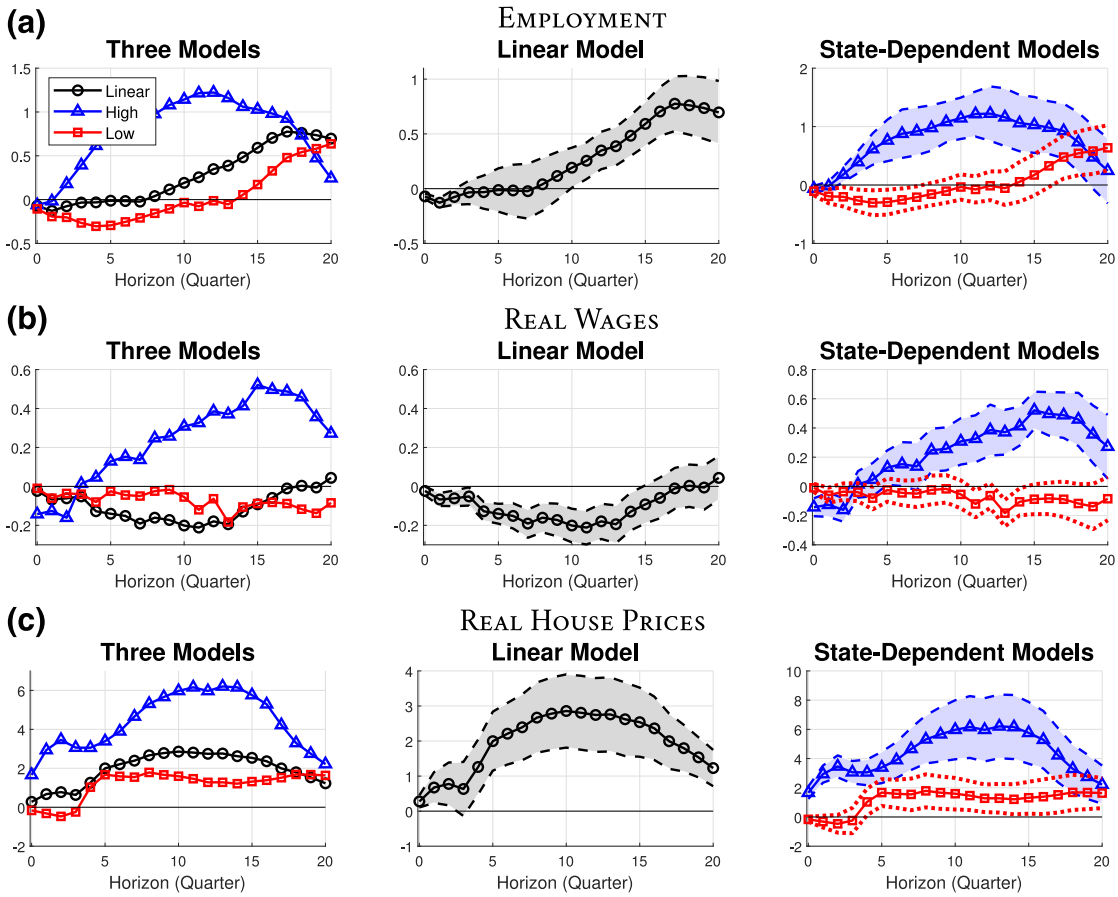

FIGURE 4 Impulse response of other key variables: Baseline specification. The impulse responses of employment, real wages, and real house prices to a 100 basis points (bps) expansionary shock for our baseline specification. The first column displays the point estimate of the impulse response of the three models: high connectedness (lines with triangles), low connectedness (lines with squares), and a linear model (lines with circles). The second column displays the impulse response and 68 percent confidence intervals for the linear model, while the third column shows the impulse response functions 68 percent confidence intervals to a monetary shock for the high- and low-connectedness states.

4 | DISCUSSION

4.1 | Mechanism

In this subsection, we examine the mechanisms driving housing market connectedness and how this connectedness amplifies the effectiveness of monetary policy. Comovements in house prices can be driven by common underlying factors, such as gradually changing credit conditions (Amromin et al., 2018; Chambers et al., 2009). When the central bank lowers interest rates, connected housing markets experience simultaneous reductions in borrowing costs, which amplify the impact of monetary policy across regions. Lower mortgage rates increase housing demand in multiple regions at once, resulting in a larger aggregate effect than would occur in isolated markets. Thus, connectedness spreads the influence of monetary policy through the housing market's credit channel, amplifying its overall impact.

Additionally, comovements may arise from the information channel (Fischer et al., 2021), where financing decisions and negotiations rely on comparable past sales prices as anchors for home values. These anchors, such as appraised values, influence both buyers' and sellers' behavior, leading past sales prices to directly and indirectly affect current sales prices (Murfin & Pratt, 2018). Comparable sales data and price anchors in connected housing markets create feedback loops. When monetary policy drives price changes in one region, those price changes are used as reference points in other regions, influencing home valuations and negotiations. This causes price shifts in one area to quickly propagate, amplifying the policy's effect on housing prices. As a result, even a localized change triggered by monetary policy can have far-reaching consequences through this information channel, amplifying the overall impact.

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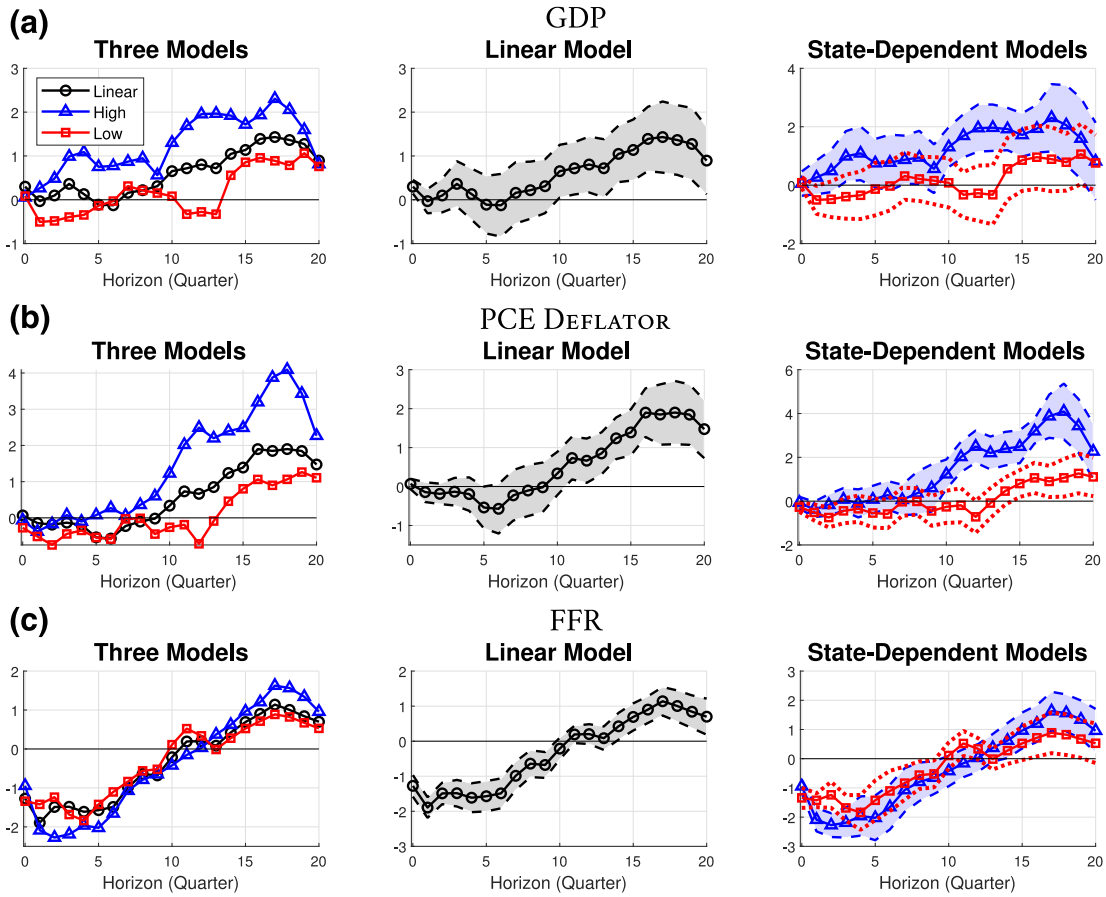

FIGURE 5 Impulse response of headline variables: Romer and Romer monetary shock. The impulse responses of GDP, the personal consumer expenditure (PCE) deflator, and the federal funds rate (FFR) to a 100 basis points (bps) expansionary Romer and Romer (2004) monetary shock. The first column displays the point estimate of the impulse response for three models: high connectedness (lines with triangles), low connectedness (lines with squares), and a linear model (lines with circles). The second column displays the impulse response and 68 percent confidence intervals for the linear model, while the third column shows the impulse response functions' 68 percent confidence intervals to a monetary shock for the high- and low-connectedness states.

Importantly, existing research suggests that synchronized bank lending activity leads to synchronized house prices. Choi and Andrew Hansz (2021) find that increased banking integration drives greater comovement in urban house prices. Similarly, Landier et al. (2017) show that house price growth correlation is strongly tied to financial integration across state pairs. Berrospide., Black and Keeton (2016) also note that interstate banking deregulation enables local economic shocks to be more easily transmitted across regions through banks' internal capital markets. Increased banking integration means that when monetary policy changes, banks across connected regions adjust their lending practices in a synchronized manner. If the central bank cuts interest rates, all regions benefit from the easier credit conditions simultaneously. This synchronized response amplifies the housing market's overall reaction, as lower borrowing costs drive up demand and house prices more uniformly across regions.

4.2 | Connectedness and local housing market dynamics

This subsection explores potential differences in the impact of monetary policy shocks on house prices across different local housing markets and how market interconnectedness affects these dynamics. Figure 8 provides a clear visualization of the response of regional house prices to a Romer and Romer (2004) monetary policy shock in the high- and low-connectedness states. The solid line represents the median response at each horizon, while the dashed lines indicate the 20th and 80th percentiles at each horizon. As illustrated in the left panel of Figure 8, local house prices tend to increase and move together in response to an expansionary monetary policy shock when regional housing markets

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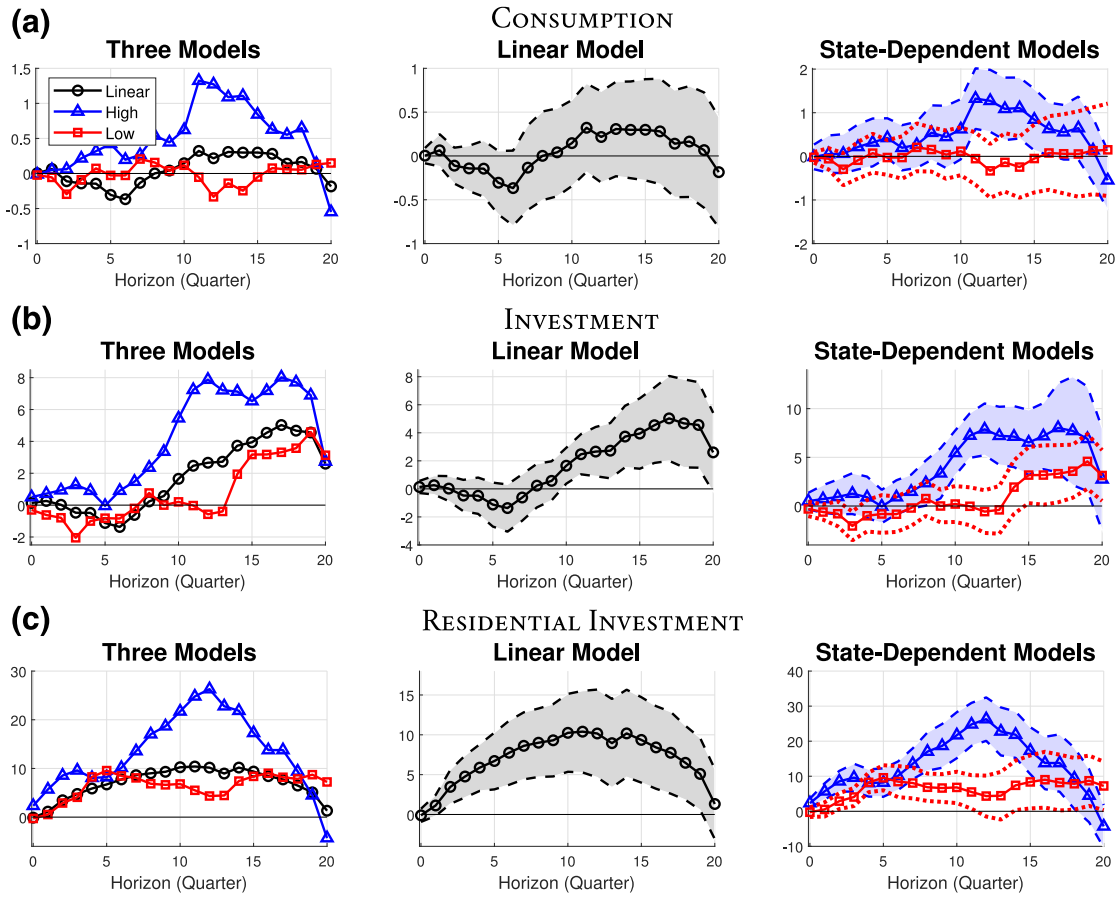

FIGURE 6 Impulse response of expenditure variables: Romer and Romer monetary shock. The impulse responses of consumption (nondurable goods and services), private fixed investment, and residential fixed investment to a 100 basis points (bps) expansionary Romer and Romer (2004) monetary shock for the three models: high connectedness (lines with triangles), low connectedness (lines with squares), and a linear model (lines with circles). The second column presents the impulse response and 68 percent confidence intervals for the linear model, while the third column shows the impulse response functions' 68 percent confidence intervals to a monetary shock for the high- and low-connectedness states.

are more synchronized. At the peak of the total house price response, which occurs 15 quarters after the shock, all states except for one show a positive response.

We also examine the correlation and heterogeneity of regional markets, finding an average cross-sectional correlation coefficient of 0.69 and an average cross-sectional coefficient of variation of around 0.7 in the high-connectedness state. In contrast, local house price responses are much more heterogeneous in the low-connectedness state. The right panel of Figure 8 demonstrates that local house prices do not move together in response to an expansionary monetary policy shock when regional housing markets are less synchronized. For instance, 18 states exhibit a negative price response 15 quarters after the shock. The average correlation is significantly lower in the low-connectedness state (0.24) than in the high-connectedness state, while the average coefficient of variation is substantially larger (1.1).[15] The results indicate that the existence of a national factor that reflects the synchronized behavior of all housing markets could be essential for enhancing the effectiveness of monetary policy in the high-connectedness state. Further elaboration on this matter will be provided in the upcoming subsection.

4.3 | Connectedness and business cycles

As previously discussed, there appears to be no discernible cyclicality in housing market connectedness, resulting in a near-zero correlation with output fluctuations (see Figure A3 in the appendix). However, it is worth exploring whether the degree of housing market connectedness affects the effectiveness of monetary policy during economic downturns. To this end, we analyze the impact of monetary shocks on states that intersect both business cycles and housing market

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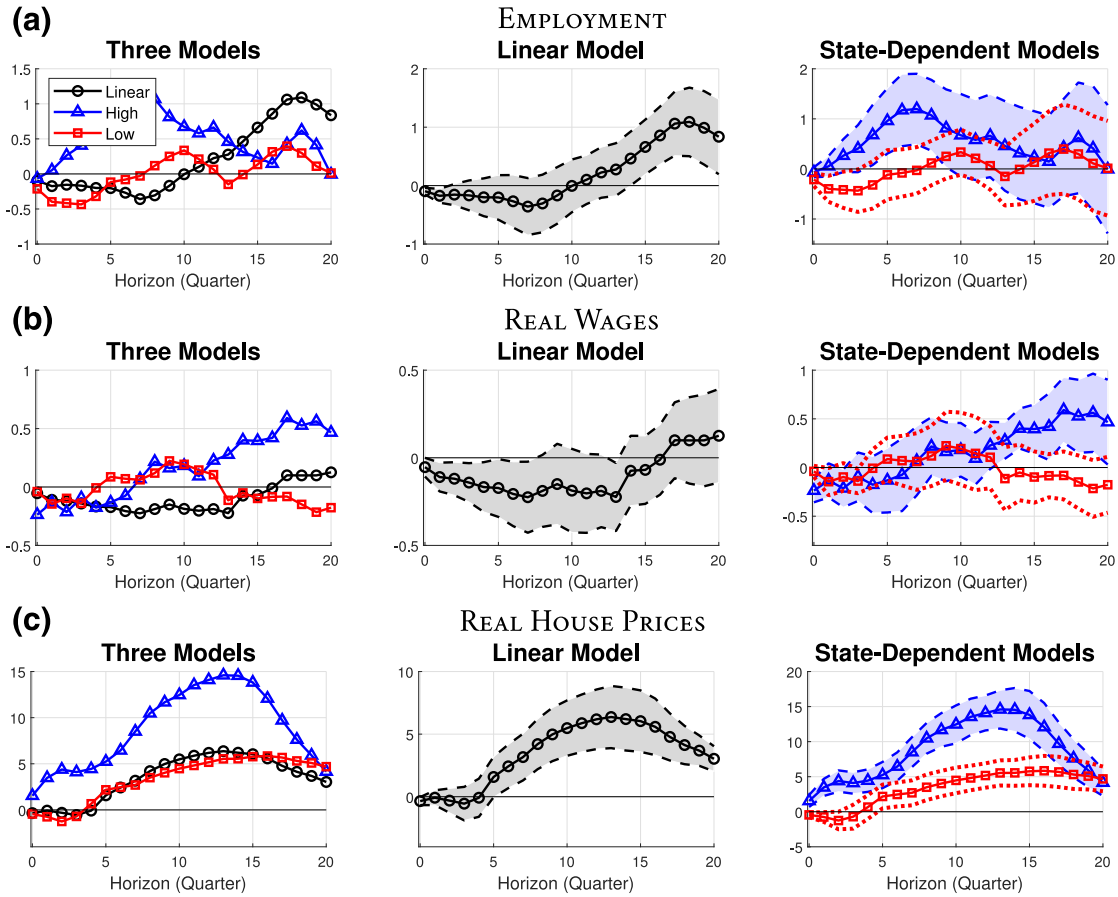

FIGURE 7 Impulse response of other key variables: Romer and Romer monetary shock. The impulse responses of employment, real wages, and real house prices to a 100 basis points (bps) expansionary Romer and Romer (2004) monetary shock. The first column displays the point estimate of the impulse response of the three models: high connectedness (lines with triangles), low connectedness (lines with squares), and a linear model (lines with circles). The second column displays the impulse response and 68 percent confidence intervals for the linear model, while the third column shows the impulse response functions 68 percent confidence intervals to a monetary shock for the high- and low-connectedness states. Still, there is evidence that the responses of employment, real wages, and real house prices are larger in the high-connectedness state, which might be suggestive evidence for stronger aggregate demand effects in the high-connectedness state.

connectedness cycles. Using quarterly GDP, we define a period as a boom state (bust state) if the deviation from the trend during the period is positive (negative).

Figure 9 illustrates the IRFs to a monetary policy shock, distinguishing between booms (the upper panel) and busts (the bottom panel), as well as high versus low levels of housing market connectedness. Overall, states characterized by a boom and high connectedness exhibit the largest responses in terms of GDP and inflation. Specifically, during a boom period, both GDP and the PCE deflator exhibit more significant responses in high-connectedness states compared to low-connectedness states. Conversely, the smallest responses occur when there is a slump coinciding with low connectedness. This indicates that monetary policy is most effective when considering high-connectedness states experiencing an expansion.

5 | SENSITIVITY ANALYSIS

In this section, we examine the robustness of our baseline results to various specifications. We consider alternative definitions for our state variable, different specifications for housing market connectedness variables, and also consider a different sample. For these sensitivity analyses, we employ the baseline identification scheme (the identification scheme under timing restrictions) for the monetary shock.

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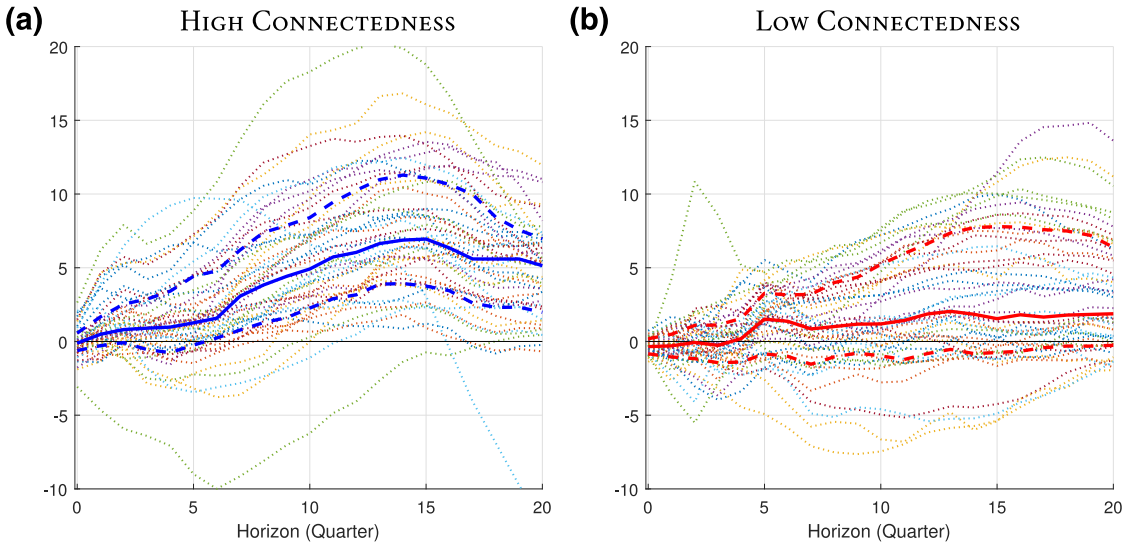

FIGURE 8 Connectedness and local housing market dynamics. The response of regional house prices to a monetary policy shock in the high- and low-connectedness states. The solid line represents the median response at each horizon, while the dashed lines indicate the 20th and 80th percentiles at each horizon.

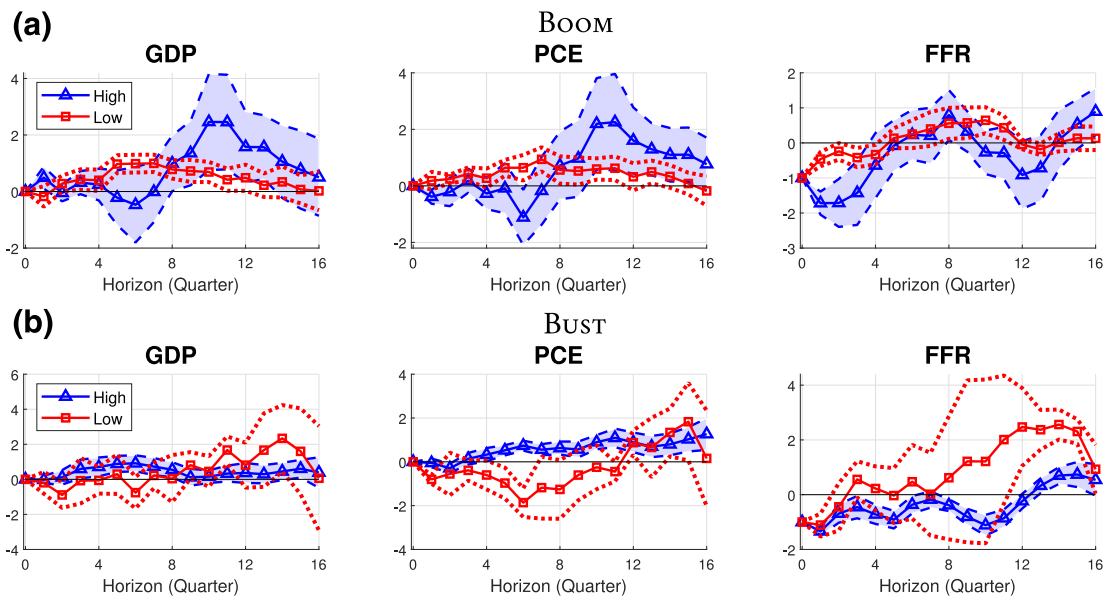

FIGURE 9 Business cycles and connectedness. The impulse responses of the headline variables to a 100 basis points (bps) expansionary monetary shock (a) during boom periods and (b) during bust periods.

5.1 | Smooth transition between states

We consider the smooth transition-local projection employed in Tenreyro and Thwaites (2016). The impulse response of variable y_t at horizon h in state $s \in \{L, H\}$ is estimated as the coefficient $\gamma_{s,h}$ in the below regression:

$$y_{t+h} = F(z_{t-1})\left[\alpha_{H,h} + \boldsymbol{\gamma_{H,h}}'\boldsymbol{x}_t + \beta_{H,h}\varepsilon_t\right] + (1 - F(z_{t-1}))\left[\alpha_{L,h} + \boldsymbol{\gamma_{L,h}}'\boldsymbol{x}_t + \beta_{L,h}\varepsilon_t\right] + u_{t+h}, \tag{4}$$

We employ a smooth increasing function of an indicator of the state of the economy, $F(z_t)$:

$$F(z_t) = \frac{\exp(\kappa z_t)}{1 + \exp(\kappa z_t)} \tag{5}$$

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where $z_t = \frac{\hat{C}_t}{\sigma_{\hat{C}}}$, and $\hat{C}_t$ and $\sigma_{\hat{C}}$ are the HP filtered connectedness and its standard deviation, respectively. The parameter κ controls the smoothness of the transition from a high-connectedness state to a low-connectedness state in the economy, affecting the intensity of regime switching. It is important to note that, as κ increases, $F(z_t)$ becomes more similar to a discrete regime-switching setup. In the smooth transition-local projection specification, the quantitative size of the connectedness can be reflected, which differs from the dummy variable used in Equation (3).

Figure 10 demonstrates the effects of monetary shocks on headline variables (GDP, the PCE deflator, and the FFR) according to connectedness states with different values of κ. As can be seen in Figure 10, the results for the three different smoothing parameters ($\kappa =$ 3, 5, and 10) are very similar to our baseline results.[16] One can see that the responses of GDP and the price level are significantly larger in the high-connectedness state, and the responses of the FFR in both states are quite similar.[17] The qualitative message of the benchmark analysis remains unchanged: the results are robust to reasonable changes in the intensity of regime switching, κ.

5.2 | Including zero lower bound period

As the FFR remained close to zero during the 2008–2015 period, we had to end our sample in 2007q4. However, it is of interest to evaluate whether our baseline results remain robust if we extend the sample period. To address this issue, we utilize the shadow FFR constructed by Wu and Xia (2016) for the subperiod 2009q1–2016q4. As a result, our extended sample period now spans from 1981q1–2016q4.

Figure 11 displays the IRFs from the linear and state-dependent models using our baseline monetary shock specification over this extended sample period. The upper panel presents the response of headline variables (GDP, PCE deflator, and shadow rate) to the monetary policy shock. Once again, we can see that the results are quite similar to the baseline results. In the linear model, output rises in response to the expansionary monetary policy. The PCE deflator's response is sticky at first, but it begins to rise after a delay. Regarding the state-dependent model, output and the price

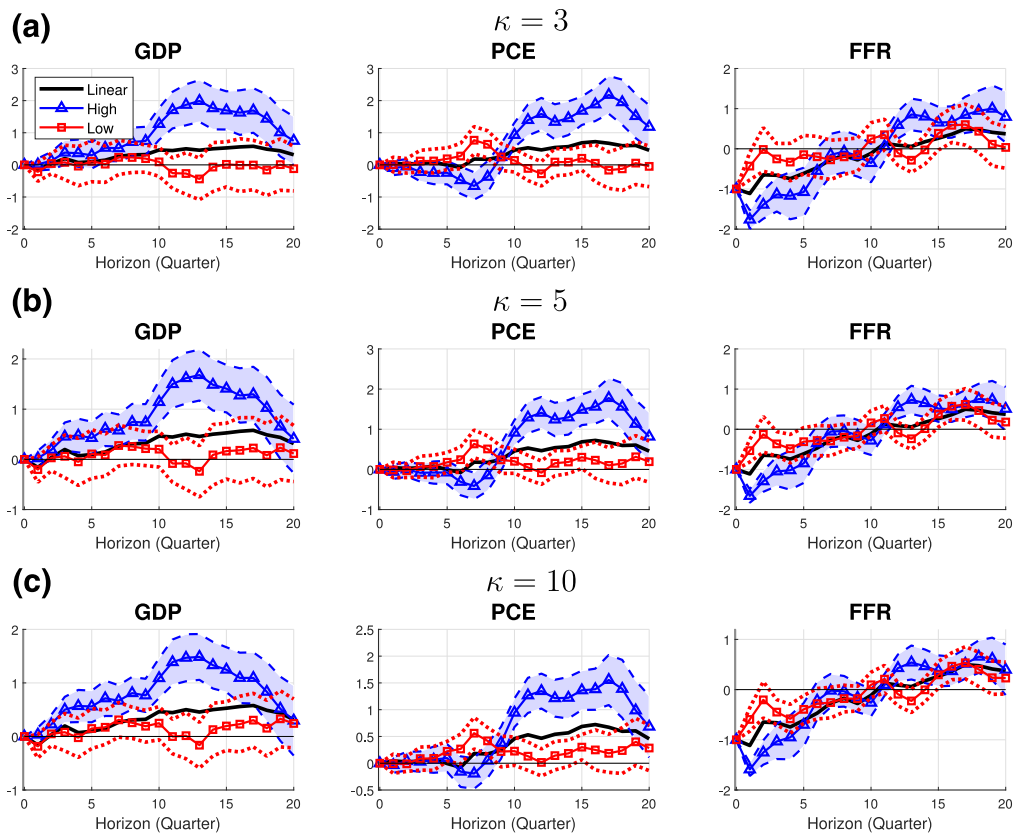

FIGURE 10 IRFs to a monetary shock with different κ. The impulse responses of GDP, the PCE Deflator, and federal funds rate (FFR) to a 100 basis points (bps) expansionary monetary shock with 68 percent confidence intervals for linear model (black lines), the high- and low-connectedness states (blue and red lines, respectively).

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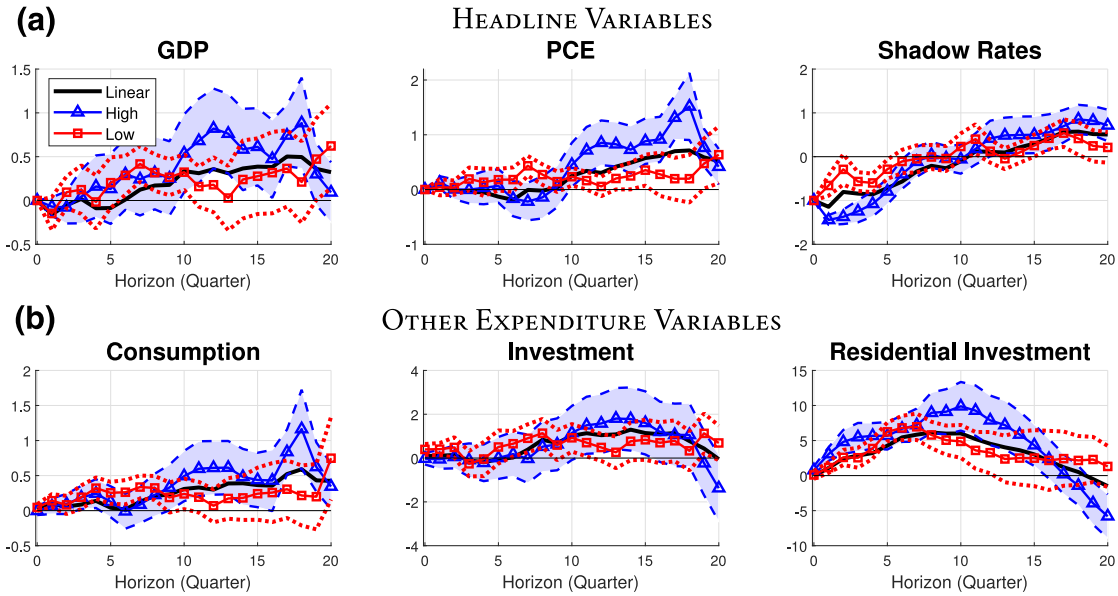

FIGURE 11 IRFs to a monetary shock: Including the Zero Lower Bound (ZLB) periods. The impulse responses of the key aggregate variables to a 100 basis points (bps) expansionary monetary shock with the extended sample period is 1981q1–2016q4.

level respond more strongly in the high-connectedness state than in the low-connectedness state. The state-dependent response of the shadow rate is similar at most horizons.

The bottom panel displays the response of consumption, investment, and residential investment for the longer sample. In the case of the linear impulse response, GDP components respond to the monetary policy shock with a persistent hump shape. Regarding the state-dependent model, the response of the three variables is more significant overall in the high-connectedness state than in the low-connectedness state, even if the differences between the two states become much smaller.[18]

5.3 | Including macro variables for connectedness measures

In the benchmark specification for estimating housing connectedness, we do not include any other macroeconomic variables. However, housing market connectedness can be influenced by common macroeconomic factors, such as interest rates. As a robustness check, we include macroeconomic and financial variables as controls in the VAR system and re-estimate the housing connectedness measure to assess its sensitivity to these controls. Specifically, we include monthly data on industrial production, the consumer price index, the unemployment rate, the federal funds effective rate (using the shadow rate for the ZLB period), stock prices, and stock market volatility.[19]

Figure 12 presents the levels of the connectedness index for both the benchmark case, where no macroeconomic controls are included, and the case where macroeconomic variables are added as controls. As expected, the overall movement of the connectedness index in both cases is very similar, indicating that the inclusion of macroeconomic factors does not drastically alter the pattern of housing market connectedness. Both measures move closely together, with little discernible difference between the two specifications, reflected in a high correlation of 0.96. This suggests that the fundamental structure of connectedness between housing markets remains robust even when controlling for these broader economic variables.

Figure 13 displays the IRFs from both the linear and state-dependent models using the new connectedness index. The upper panel presents the responses of headline variables—GDP, PCE deflator, and FFR—to a monetary policy shock. The results are largely consistent with the baseline specification. In the linear model, output increases following an expansionary monetary policy shock, while the PCE deflator shows an initially sticky response before gradually rising. In the state-dependent model, output and the price level exhibit a stronger response in the high-connectedness state compared to the low-connectedness state. The response of FFR is relatively similar across both states at most horizons. The bottom panel illustrates the responses of consumption, investment, and residential investment. Overall, the reactions of these three variables are more pronounced in the high-connectedness state compared to the low-

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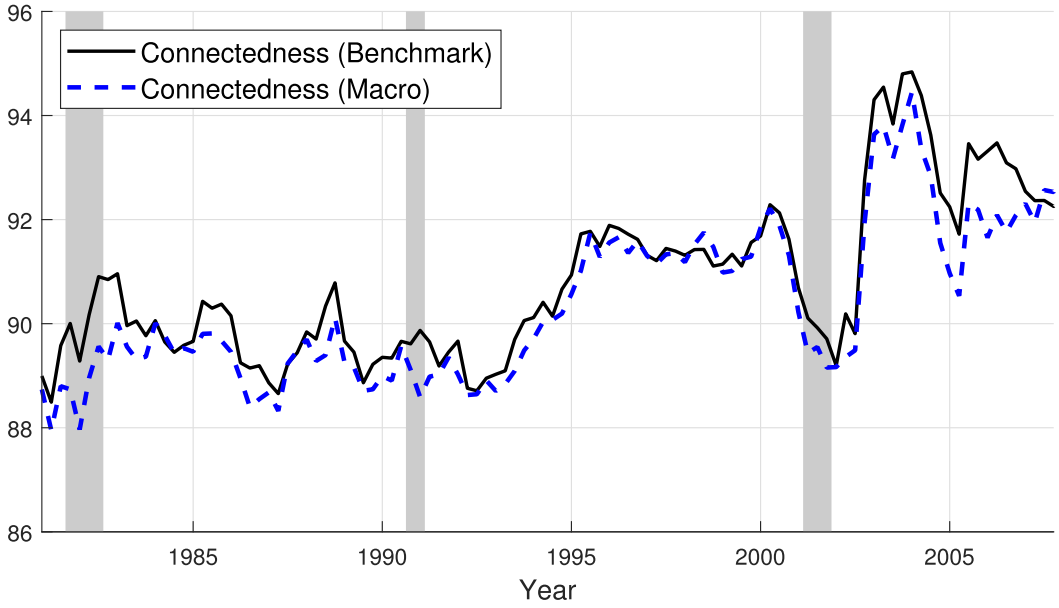

FIGURE 12 Connectedness index: Trend and cyclical components. The levels of the connectedness index are shown for both the benchmark case and the case with macroeconomic controls. The gray-shaded regions indicate NBER recession periods.

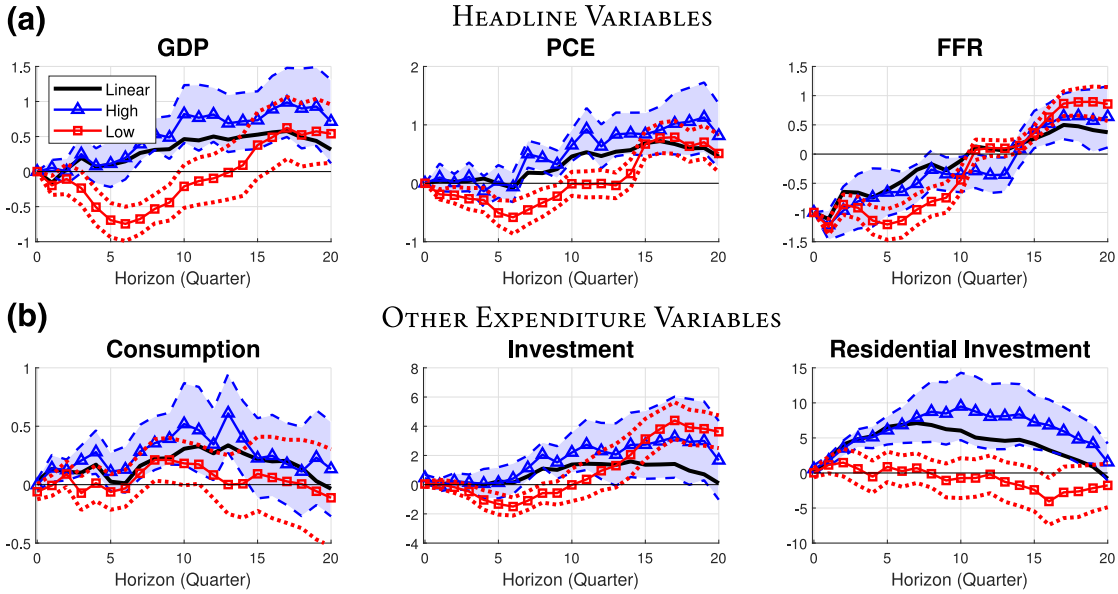

FIGURE 13 IRFs to a monetary shock: Macro controls for connectedness measures. The impulse responses of the key aggregate variables to a 100 basis points (bps) expansionary monetary shock.

connectedness state. While not shown in the figure, we also find that other macro variables of interest also exhibit stronger responses in the high-connectedness state than in the low-connectedness state.

5.4 | Horizon, window size, and lags

As the connectedness measures used in this paper are sensitive to the choice of forecast horizon, we explore whether our baseline results are robust to different forecast horizons, rolling window sizes, and lag choices. As shown in Figure A4 in the appendix, we find that the overall qualitative picture from the IRFs remains unchanged.[20]

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6 | CONCLUSION

This paper examines the impact of housing market connectedness on the monetary transmission mechanism in the U.S. economy. Through the use of state-dependent local projection methods, our study reveals that the efficacy of monetary policy is more pronounced in high-connectedness states. Specifically, we observe that the response of key macroeconomic variables, such as GDP, consumption, residential fixed investment, employment, and house prices, to a monetary policy shock is significantly greater when the initial level of total connectedness is high compared to its long-term trend. These findings are not only robust to alternative identification methods of monetary policy shocks but also various modifications to the empirical model. Our study also highlights the importance of considering the role of national factors in the transmission of monetary policy. Specifically, our results suggest that the strengthening of the national factor channel may play a crucial role in generating more effective monetary policy in high-connectedness states. We also discovered a compelling result in our analysis of the effects of monetary shocks on states that intersect both business cycles and housing market connectedness cycles, namely that monetary policy is most effective when high-connectedness states experience an expansion.

The importance of this paper lies in offering valuable insights into whether the spillover effect among neighboring housing markets amplifies or dampens the effects of monetary policy. Changes in interest rates can impact the housing market in various ways, including lowering mortgage rates, which can ease the financing process for potential homebuyers. Moreover, increased demand for housing in one region can spread to other regions, leading to an overall rise in aggregate demand. This paper suggests that this effect can be particularly significant in a synchronized housing market.

These results have implications for stabilization policy designs. If monetary policy changes have a limited impact in the low-connectedness state, policymakers may need to rely on unconventional monetary policies or other measures, such as fiscal or financial policies, to achieve the desired expansionary effect. Additionally, the findings call for macroeconomic models that generate a higher sensitivity in the response of the economy depending on connectedness among local markets.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in OPENICPSR at https://doi.org/10.3886/E216361V3 (Lee & Ma, 2025).

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ENDNOTES

[1] We find that these episodes of low or high connectedness do not necessarily align with business cycles.

[2] For example, greater housing market synchronization, driven by a stronger national factor, increases systemic risk at the national level due to interconnectedness, which in turn reduces the effectiveness of policy interventions.

[3] The concept of spatial transmission has been pondered upon in alternative contexts as well. Notably, Giroud and Mueller (2019), along with Bahaj et al. (2022), explore intriguing spatial mechanisms related to housing.

[4] α can be selected by cross validation, but it substantially increases computational burdens. It should be also noted that the elastic net estimators are consistent as long as $\alpha \in (0, 1)$, that is, both the ridge and lasso penalty weights are positive.

[5] As is well-known in the literature, Cholesky factorization is not robust to variable ordering, which can potentially cause a problem when trying to obtain both system-wide and directional connectedness measures.

[6] We also include deterministic trends (linear and quadratic time trends).

[7] L denotes low connectedness while H denotes high connectedness.

[8] The main results remain robust when excluding Alaska, Hawaii, and Washington, D.C.

[9] In a later section, we investigate the robustness of our results to each of these choices.

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[10] Our sample ends in Q4 2007 to avoid the Zero Lower Bound (ZLB) period on the federal funds rate. We will expand our sample period later to see if the results remain robust when recent periods are included (see Figure A2 in the appendix for the connectedness index over the extended period).

[11] For a depiction of the specific comovement between the cyclical components of the connectedness index and GDP, please refer to Figure A3 in the appendix.

[12] The distribution of these monetary shocks (e.g., the signs and sizes) looks similar across high- and low-connectedness states.

[13] Our results are robust from a qualitative perspective, even when we include durable goods in consumption.

[14] Just like we did for the baseline specification, we conduct statistical tests to compare the cumulative impulse response functions for the level of GDP and inflation in the high-connectedness state with those in the low-connectedness state. Our findings indicate that the cumulative effects on the level of GDP and inflation are significantly greater in the high-connectedness state than in the low-connectedness state, particularly over longer horizons. However, there is no statistically significant difference in the cumulative response of the federal funds rate between the two states at most horizons.

[15] Additionally, it should be noted that the median response of house prices is greater in the high-connectedness state compared to the low-connectedness state, which is in line with the findings of our analysis at the aggregate level (Figure 4).

[16] Tenreyro and Thwaites (2016) set $\kappa = 3$ and consider a wide range of parameters for the intensity of regime switching from 1 to 10.

[17] Our statistical analyses reveal that, for three specifications, the high-connectedness state has greater cumulative impacts on GDP and inflation than the low-connectedness state, especially over extended periods at a significance level of 10%.

[18] Although not depicted in this figure, we also find that employment, real wages, and house prices have a significantly greater response in the high-connectedness state than in the low-connectedness state.

[19] Since Economic Policy Uncertainty (EPU) and the CBOE Volatility Index (VIX) are not available for our entire sample, stock market volatility is estimated using an AR(1)-GARCH(1,1) model. All variables are either in log differences or first differences to ensure stationarity.

[20] In some cases, the response of GDP or inflation is not statistically different across regimes but similar in qualitative terms.

[21] We then convert monthly *R*-squared measures to quarterly ones.

[22] Del Negro and Otrok (2007) that the role of monetary policy in determining house price dynamics is limited, suggesting that there may be other factors at play. For example, financial integration is one potential factor that may explain the existence of a global factor affecting house prices. According to Landier et al. (2017) and Choi and Andrew Hansz (2021), the degree of financial integration is strongly correlated with the growth of house prices.

[23] As found in Del Negro and Otrok (2007), we also find that the national factor's strength has significantly risen since 2000 in comparison to the periods preceding 2000.

[24] To compare our baseline results, we set the smoothing to be 1600 and use sample periods form 1981:q1–2007:q4.

[25] Our statistical tests show that the high-connectedness state has larger cumulative effects on GDP and inflation than the low-connectedness state over longer horizons at the 5% significance level. However, there is no difference in the cumulative response of the federal funds rate between the two states.

[26] We also find that employment, real wages, and real house prices respond more strongly in the high-strength state.

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APPENDIX A

A.1 | Data Sources

This subsection explains the data sources used in the paper. All data, except for regional house prices and the shadow rate, are drawn from the St. Louis Federal Reserve Economic Data.

A.1.1 | Federal Reserve Economic Data (FRED)

- GDP: Real Gross Domestic Product; PCE deflator: Personal consumption expenditures (implicit price deflator) index 2012 = 100; FFR: Federal funds effective rates; Consumption: The sum of nondurable goods and services; Investment: The sum of private residential fixed Investment and nonresidential fixed investment; Employment: All employees in the nonfarm sector; Real wage: Average hourly earnings of production and nonsupervisory employees in the private sector.

A.1.2 | Other sources

- **Freddie Mac**
  - National house price; Regional house price for 50 states and the District of Columbia.
- **Federal Reserve Bank of Atlanta**
  - Shadow rate: the Wu-Xia Shadow FFR.

A.2 | Connectedness and national factor

In this subsection, we consider a possible explanation for why monetary policy has a large stimulus effect in the high-connectedness state. A natural candidate is a national factor that captures the common movements in all housing markets. In other words, housing markets are more synchronized if a single national factor is the main driver of house price fluctuations rather than idiosyncratic regional-specific factors. To quantify this relationship, we estimate a national factor using the 51 housing price return series and then see if monetary policy is more expansionary when the strength of a national factor is relatively high. We take three steps to estimate the average strength of the national factor.

Step 1: We estimate the national factor series using the 51 housing price return series $(R_{i,t})$ for the sample period of 1976:m1–2020:m12. To determine the national factor series, we use the eigenvalue ratio criteria developed in Ahn and Horenstein (2013), and we assume that the first factor represents the national factor (GF_t).

Step 2: Given the factors estimated from Step 1, we apply to 120-month rolling regression method to compute time-varying R-squared measures. Specifically, we regress the ith housing return series on GF_t:

$$R_{i,t} = \alpha_i + \beta_i GF_t + \varepsilon_{i,t} \tag{A1}$$

Step 3: We average the 51 R-square values measured at each window point,[21] which dented by Υ_t.

Υ_t can capture the overall strength of the national factor, GF_t, relative to the regional-specific factors, $\varepsilon_{i,t}$. In principle, individual housing price returns are more synchronized when Υ_t is high. We find that the strength of the national

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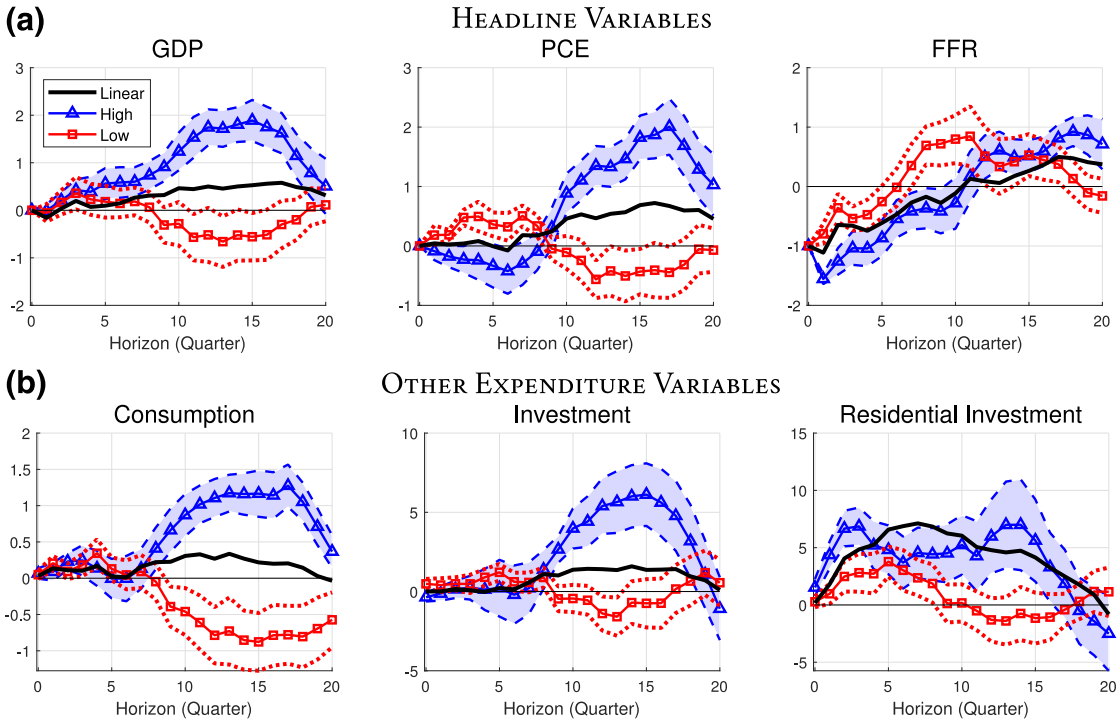

FIGURE A1 IRFs to a monetary shock between low- and high-strength states. The impulse responses of the headline variables and expenditure variables to a 100 basis points (bps) expansionary monetary shock: high strength (lines with triangles), low strength (lines with squares), and a linear model (solid lines).

factor, Υ_t, corresponds well to our housing connectedness measure, showing a high correlation of 0.7.[22] This result is in line with the observation made by Del Negro and Otrok (2007) that the rise in house prices before the Great Recession was mainly driven by the national component rather than the local one.[23] We next investigate whether monetary policy is more powerful with higher Υ_t.

We examine whether the transmission of monetary shocks is state-dependent on the strength of the national factor, Υ_t. As in the benchmark specification, we detrend Υ_t using the HP filter and separate it into low- and high-strength states.[24] Figure A1 provides insightful analysis of the impact of state-dependent models on the strength of the national factor as a different state specification. The results demonstrate that the response of output and the price level is significantly greater when the national factor is the primary driver of regional housing markets. In terms of GDP components, the second row of Figure A1 highlights a noticeable difference between high- and low-strength states.[25] Notably, consumption, investment, and residential investment display a much stronger response in the high-strength state than in the low-strength one.[26] The results presented in the study provide compelling evidence that the strengthening of the national factor channel could potentially explain the observed increase in monetary policy effectiveness in the high-connectedness state.

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A.3 | Additional figures

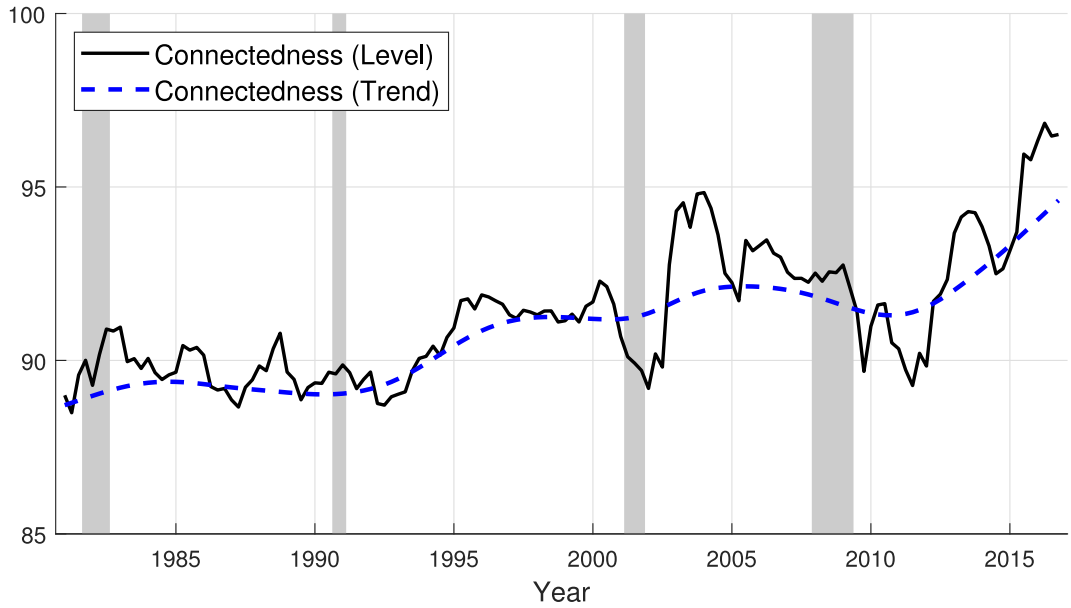

FIGURE A2 Connectedness index: Trend and cyclical components until 2016. The trend and level of the connectedness index until 2016. The gray-shaded regions indicate NBER recession periods.

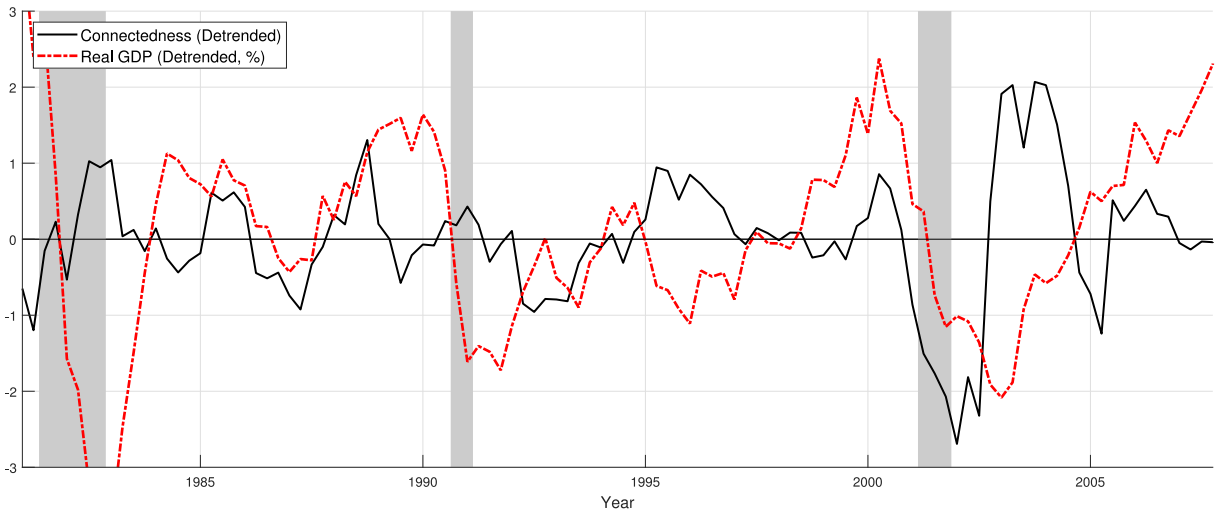

FIGURE A3 Detrended connectedness and GDP. The detrended connectedness index and GDP. The gray-shaded regions indicate NBER recession periods.

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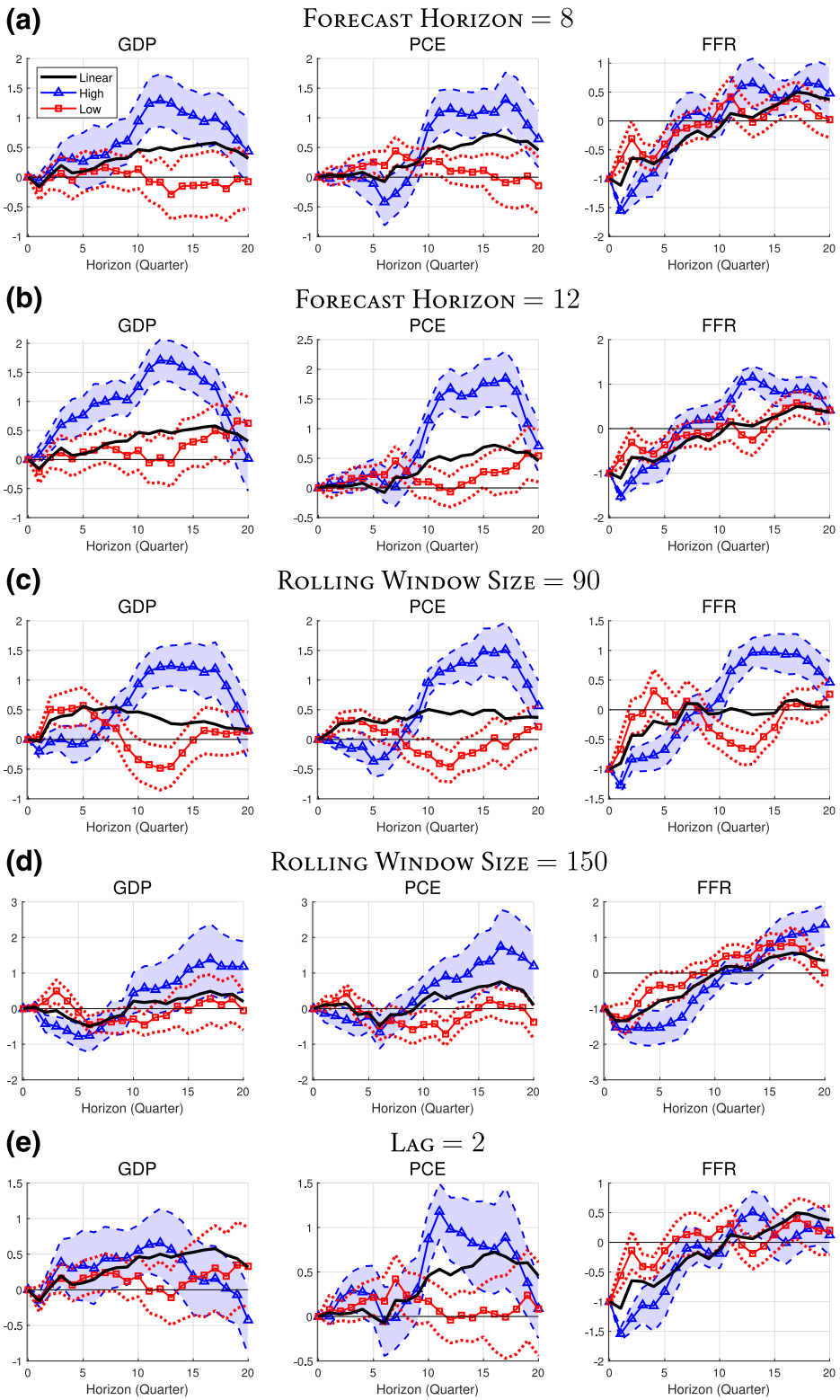

FIGURE A4 IRFs to a monetary shock with different horizons, windows, and lags. The impulse responses of GDP, the PCE Deflator, and federal funds rate (FFR) to a 100 basis points (bps) expansionary monetary shock with 68 percent confidence intervals for linear model (black lines), the high- and low-connectedness states (blue and red lines, respectively).

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